Practice Set A

Class 12 Mathematics, Chapter 13: Probability

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, Board-pattern — 1 mark

If $P(A|B) = 0.3$, $P(A) = 0.4$ and $P(B) = 0.8$, then $P(B|A)$ is:

(A) 0.6 **(B)** 0.3 **(C)** 0.06 **(D)** 0.4

2 MCQ, PYQ-style — 1 mark

If E and F are two independent events such that $P(E) = \frac{2}{3}$ and $P(F) = \frac{3}{7}$, then $P(E|\bar{F})$ is equal to:

(A) $\frac{1}{6}$ **(B)** $\frac{1}{2}$ **(C)** $\frac{2}{3}$ **(D)** $\frac{7}{9}$

3 MCQ, CBSE SQP 2025-26 — 1 mark

A person observed the first 4 digits of your 6-digit PIN. What is the probability that the person can guess your PIN?

(A) $\frac{1}{81}$ **(B)** $\frac{1}{100}$ **(C)** $\frac{1}{90}$ **(D)** 1

4 MCQ, CBSE SQP 2024-25 — 1 mark

For any two events A and B, if $P(\bar{A}) = \frac{1}{2}$, $P(\bar{B}) = \frac{2}{3}$ and $P(A \cap B) = \frac{1}{4}$, then $P(\bar{A}|\bar{B})$ equals:

(A) $\frac{3}{8}$ **(B)** $\frac{8}{9}$ **(C)** $\frac{5}{8}$ **(D)** $\frac{1}{4}$

5 MCQ, Board-pattern — 1 mark

A random variable X has the probability distribution X: $-1,\ 0,\ 1,\ 2$ with $P(X)$: $0.2,\ k,\ 2k,\ 0.2$. The value of k is:

(A) 0.1 **(B)** 0.2 **(C)** 0.3 **(D)** 0.6

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** Events A and B with $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cup B) = 0.8$ are independent. **Reason (R):** Two events A and B are independent if and only if $P(A \cap B) = P(A)\,P(B)$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, Board-pattern A box of oranges is inspected by examining three randomly selected oranges drawn without replacement. If all the three oranges are good, the box is approved for sale, otherwise it is rejected. Find the probability that a box containing 15 oranges, out of which 12 are good and 3 are bad, will be approved for sale.	2 marks

Level 2

18 marks

8	Short answer, PYQ-style A bank offers loans to its customers on three types of interest: fixed rate, floating rate and variable rate. From past data, a customer avails a loan on fixed rate, floating rate and variable rate with probabilities 10%, 20% and 70% respectively. The probability that a customer defaults on repayment is 5%, 3% and 1% for fixed, floating and variable rate loans respectively. A customer is found to have defaulted on loan repayment. Find the probability that he availed the loan at a variable rate of interest.	3 marks
9	Short answer, Board-pattern A fair die is thrown twice. Find the probability distribution of the number of sixes. Also find its mean.	2 marks
10	Short answer, CBSE SQP 2024-25 The probability that it rains today is 0.4. If it rains today, the probability that it will rain tomorrow is 0.8. If it does not rain today, the probability that it will rain tomorrow is 0.7. Let P_1 denote the probability that it does not rain today, P_2 the probability that it will not rain tomorrow if it rains today, P_3 the probability that it will rain tomorrow if it does not rain today, and P_4 the probability that it will not rain tomorrow if it does not rain today. **(i)** Find the value of $P_1 \times P_4 - P_2 \times P_3$. — 2 marks **(ii)** Calculate the probability of raining tomorrow. — 1 mark	3 marks
11	Short answer, CBSE SQP 2025-26 Two students Mehul and Rashi are seeking admission in a college. The probability that Mehul is selected is 0.4 and the probability of selection of exactly one of them is 0.5. Chances of selection of them are independent of each other. Find the chances of selection of Rashi. Also find the probability of selection of at least one of them.	3 marks

Q No.	Question	Marks
12	Short answer, Board-pattern A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?	2 marks
13	Short answer, CBSE SQP 2024-25 A random variable X can take all non-negative integral values and the probability that X takes the value r is proportional to 5^{-r}. Find $P(X < 3)$.	3 marks
14	Short answer, Board-pattern A die is thrown once. Let E: "the number is even" and F: "the number is a prime number". Are E and F independent? Justify.	2 marks

Level 3

25 marks

15	Long answer, PYQ-style In an office three employees Jayant, Sonia and Oliver process incoming copies of a certain form. Jayant processes 50% of the forms, Sonia processes 20% and Oliver the remaining 30% of the forms. Jayant has an error rate of 0.06, Sonia has an error rate of 0.04 and Oliver has an error rate of 0.03. The manager selects a form at random from the day's output of processed forms. If the form selected has an error, find the probability that the form is not processed by Jayant.	5 marks
16	Short answer, Board-pattern A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.	3 marks
17	Long answer, Board-pattern Two numbers are selected at random (without replacement) from the first six positive integers. Let X denote the larger of the two numbers obtained. Find the probability distribution of X and hence find its mean $E(X)$.	5 marks
18	Short answer, Board-pattern A random variable X has the probability distribution X: $0, 1, 2, 3, 4$ with $P(X)$: $k,\ k,\ 2k^2,\ 3k^2,\ 5k^2 + k$. Find k, $P(X < 2)$ and $P(X \geq 3)$.	3 marks
19	Long answer, Board-pattern Assume that the chances of a patient having a heart attack are 40%. A meditation and yoga course reduces the risk of heart attack by 30%, and prescription of a certain drug reduces its chance by 25%. At a time, a patient can choose any one of the two options with equal probabilities. It is given that after going through one of the two options, a patient selected at random suffers a heart attack. Find the probability that the patient followed a course of meditation and yoga.	5 marks

Q No.	Question	Marks
20	Case study, CBSE SQP 2025-26 **Case Study:** Excessive use of screens can result in vision problems, obesity, sleep disorders, anxiety and low retention problems. In a class of students of the age group 14 to 17, the students were put into three groups from a feedback form. The first group spent more than 4 hours per day on mobile or gaming screens, the second group spent 2 to 4 hours per day and the third group spent less than 2 hours per day. The first group (high screen time) is 60% of all the students, the second group (moderate screen time) is 30% and the third group (low screen time) is only 10%. It was observed that 80% of the first group, 70% of the second group and 30% of the third group faced severe anxiety and low retention issues. **(i)** What is the total percentage of students who suffer from anxiety and low retention issues in the class? 2 marks **(ii)** A student is selected at random, and he is found to suffer from anxiety and low retention issues. What is the probability that he/she spends screen time of more than 4 hours per day? 2 marks	4 marks

Answer key

Chapter 13: Probability, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) 0.6
2	(C) $\frac{2}{3}$
3	(B) $\frac{1}{100}$
4	(C) $\frac{5}{8}$
5	(B) 0.2
6	(D) A is false, but R is true
7	$\frac{44}{91}$

Level 2

8	$P(E_3 \mid A) = \frac{7}{18}$
9	X: $0, 1, 2$ with $P(X)$: $\frac{25}{36}, \frac{10}{36}, \frac{1}{36}$; mean $= \frac{1}{3}$
10	(i) 0.04 (ii) 0.74
11	$P(R) = 0.5$; $P(\text{at least one}) = 0.7$
12	$\frac{2}{5}$
13	$P(X < 3) = \frac{124}{125}$ (with $k = \frac{4}{5}$)
14	No, $P(E \cap F) = \frac{1}{6} \neq \frac{1}{4} = P(E)\,P(F)$, so E and F are not independent

Level 3

15	$P(\text{not Jayant} \mid E) = \frac{17}{47}$
16	$P(E_1 \mid A) = \frac{3}{8}$
17	X: $2, 3, 4, 5, 6$ with $P(X)$: $\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15}$; $E(X) = \frac{14}{3}$
18	$k = \frac{1}{5}$, $P(X < 2) = \frac{2}{5}$, $P(X \geq 3) = \frac{13}{25}$
19	$P(E_1 \mid A) = \frac{14}{29}$
20	(i) 72% (ii) $\frac{2}{3}$